Week 13 Reading: Weather generators

Stochastic weather generators and the non-homogeneous hidden Markov model

2026-11-16

Draft. This page is unfinished and will change.

Assigned

1. Greene et al. (2011).

Background

1. Section 3.5.2 of Ghil et al. (2011), the paragraph on stochastic weather generators.

Further reading

1. Steinschneider et al. (2019), §3.1-3.3.
2. Kunkel et al. (2020).

Bibliography

Ghil, M., Yiou, P., Hallegatte, S., Malamud, B. D., Naveau, P., Soloviev, A., et al. (2011). Extreme Events: Dynamics, Statistics and Prediction. *Nonlinear Processes in Geophysics*, *18*(3), 295–350. https://doi.org/10/fvzxvv

Greene, A. M., Robertson, A. W., Smyth, P., & Triglia, S. (2011). Downscaling Projections of Indian Monsoon Rainfall Using a Non-homogeneous Hidden Markov Model. *Quart J Royal Meteoro Soc*, *137*(655), 347–359. https://doi.org/10.1002/qj.788

Kunkel, K. E., Stevens, S. E., Stevens, L. E., & Karl, T. R. (2020). Observed Climatological Relationships of Extreme Daily Precipitation Events with Precipitable Water and Vertical Velocity in the Contiguous United States. *Geophysical Research Letters*, *47*(12), e2019GL086721. https://doi.org/10.1029/2019GL086721

Steinschneider, S., Ray, P., Rahat, S. H., & Kucharski, J. (2019). A Weather-Regime-Based Stochastic Weather Generator for Climate Vulnerability Assessments of Water Systems in the Western United States. *Water Resources Research*, *55*(8), 6923–6945. https://doi.org/10.1029/2018WR024446